Pandas DataFrames

Data with Dylan | Guided Notes | Python Fundamentals #7

Name: ________________________________

Date: ________________________________

☑ **HOW TO USE THESE NOTES:** Watch the video and fill in the blanks as Dylan explains each concept. Use the writing boxes to capture your own notes and examples. Complete the practice problem at the end!

Section 1: Introduction to Pandas

Topic: Using a library to analyze data

What is Pandas?

Complete the definition:

Pandas is a Python ________________________________ commonly used in the ________________________________ field.

Pandas helps you analyze data through a ________________________________.

A DataFrame is Pandas's core ________________________________ structure for storing and manipulating data.

A library is a collection of prewritten ________________________________ that you can import into a Python program.

Why use Pandas?

Pandas can make a data task more ________________________________ by reducing the number of lines of code needed.

In your own words - why might a data scientist prefer working with a DataFrame instead of manually organizing every value?

Section 2: Importing Pandas with an Alias

Topic: Loading a library with shorthand syntax

Importing the library

Complete the code below:

```
import pandas as _______
```

The word as creates an ______________________________.

An alias is a shorter ______________________________ used to refer to a longer library name.

The standard alias for Pandas is ______________________________.

Instead of repeatedly writing pandas, programmers can write ______________________________.

Why use an alias?

In your own words - why do data scientists commonly write import pandas as pd instead of only import pandas?

Write the complete import statement:

```
import ______________________________________________________________________
```

Section 3: Organizing College Data in a Dictionary

Topic: Preparing column-like data before conversion

Source data structure

Complete the source dictionary below:

```
college_data = {
    "university": [_______, _______, _______],
    "instate_tuition": [_______, _______, _______],
    "acceptance_rate": [_______, _______, _______]
}
```

Each dictionary ________________________________ represents a different type of college information.

Each dictionary value is a ________________________________ containing multiple values.

Each individual element in a list represents one ________________________________ in the final table.

In the video, each list contains ________________________________ values, so the DataFrame displays ________________________________ rows.

From dictionary structure to table structure

A dictionary key such as "university" becomes a DataFrame ________________________________.

Values in the same list position across all lists become one DataFrame ________________________________.

In your own words - explain how Python turns dictionary keys and list values into a spreadsheet-like table.

Section 4: Creating a Pandas DataFrame

Topic: Converting the dictionary into tabular data

Initialize the DataFrame

Complete the code below:

```
df = pd.______________________________(college_data)
```

df is a common shorthand variable name for a ______________________________.

The DataFrame class receives the ______________________________ you want to convert.

The constructor name is case-sensitive: DataFrame uses a capital ______________________________ and a capital ______________________________.

Data shape checkpoint

For the dictionary to form a clean table, each list should contain the same number of ______________________________.

A mismatch in list lengths can lead to a ______________________________.

Circle the VALID DataFrame initialization:

Choice A	Choice B	Choice C
`df = pd.DataFrame( college_data)`	`df = pd.dataframe( college_data)`	`df = pandas.dataframe( college_data)`

Why are the other choices incorrect?

Section 5: Reading the DataFrame

Topic: Interpreting rows, columns, and previews

Looking at the result

After creating the DataFrame, enter:

```
df
```

Pandas automatically recognizes dictionary keys as ______________________________.

Pandas treats matching elements from each list as a separate ______________________________.

The first value from every list forms the ______________________________ row.

Because the example contains only three rows, displaying df is enough to view the complete dataset.

Previewing data

df.head() is useful when you want to see a ______________________________ of a larger DataFrame.

In your own words - what does it mean that corresponding list positions become rows?

Pandas DataFrame Reference

CODE	METHOD / ATTRIBUTE NAME	RESULT / NOTES
`pd.DataFrame(college_data)`	______________________________	Converts the dictionary into a DataFrame.
`df`	______________________________	Displays the created DataFrame.
`df.head()`	______________________________	Shows a preview of the DataFrame's top rows.
`df.to_csv(...)`	______________________________	Saves DataFrame data into a CSV file.

Section 6: Preparing to Export a DataFrame

Topic: Saving structured data as a CSV file

Exporting the DataFrame

A CSV file stores data in a structured, spreadsheet-friendly format.

Complete the method below:

```
df.______________________________("college_data.csv")
```

The export operation can be completed in _______________________________ line of code.

The DataFrame variable must exist before you can export it.

Connecting this lesson to File I/O

The dictionary is first converted into a _______________________________.

The DataFrame can then be saved as a _______________________________ file.

In your own words - why is it useful to convert data into a DataFrame before exporting it?

Before exporting

Write the library installation command that you may need before importing Pandas:

```
________________________________________________________________________________
```

Section 7: Putting It All Together

Topic: Dictionary-to-DataFrame reference table

Using what you have learned, complete the full structure comparison table:

DICTIONARY COMPONENT	DATAFRAME COMPONENT	STRUCTURAL LAYOUT DESCRIPTION	YOUR OWN EXAMPLE
college_data dictionary	Complete DataFrame	The full collection of related data becomes one spreadsheet-like table.	
Dictionary key	Column name	A key such as "university" labels one vertical column.	
List value	Column values	Every item in one list is placed vertically within its matching column.	
Matching list position	Row	Values at the same position across lists become one record.	
First element from each list	First row	The first university, tuition value, and acceptance rate are displayed together.	

Section 8: Quick Recap

Test yourself - try to complete these without looking back!

1. The standard shorthand alias in import pandas as pd is ______________________________.

2. A DataFrame is Pandas's core ______________________________ structure for storing and manipulating data.

3. To create a DataFrame from college_data, write pd.______________________________(college_data).

4. The class name must be written as DataFrame, with capital ______________________________ and ______________________________ letters.

5. Dictionary keys become DataFrame ______________________________, while matching list elements become DataFrame ______________________________.

Practice Problem

⚡ **Challenge - Pause the video and try this yourself before Dylan reveals the answer!**

```
df = pd.DataFrame(college_data)
```

Goal: Permanently export the newly created DataFrame df into a structured CSV file named college_data.csv.

Why? Why should .to_csv() be called on df instead of directly on college_data?

Why? Why should the filename end with .csv?

Write your full statement below:

```
df.________________________________________________
```

How did it go? What part tripped you up?

Personal Notes

Use this space to capture any additional observations, questions, library installation commands (pip install pandas), or ideas from the video.

Key Terms Reference

Use this table to build your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Pandas	
DataFrame	
Library	
Alias	
pd	
import	
as Keyword	
Dictionary	
Dictionary Key	
List Value	

Key Terms Reference (continued)

Continue building your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Column	
Row	
df	
pd.DataFrame()	
Case-Sensitivity	
ValueError	
.head()	
.to_csv()	
CSV File	
pip install pandas	